

GreenCommute

A worker commuter app for GreenTex Apparel, Gazipur

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Figma design file	figma.com/design/aWSZZboavo5blt92qFQnsz
Prototype · boarding flow	Play from Language → Sign-in → Home → Door code → Ticket → In transit
Prototype · SOS trigger	Play from In transit → SOS → countdown → delivered
Screens page (all frames)	03 Screens
Process repository	github.com/Tayebbbb/ui-ux-carnival — full phase documents, decision log, exports

Figma pages: [00 Cover](#) · [01 Flow & Sitemap](#) · [02 Wireframes](#) · [03 Screens](#) · [04 Design System](#) · [05 Accessibility](#) · [06 Prototype](#) · [07 Redlines & Handoff](#) · [Archive](#)

Evidence tags used throughout: **[E-01]** Anisur's email · **[E-02]** kick-off notes · **[E-03]** Monira Khatun (HR) voice note · **[E-04]** SMS · **[I]** inference · **[A]** assumption, unverified · **[D]** design decision. Nothing in this document is invented research; where the pack is silent it is marked [A].

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7. Tools, AI use & credits

1 · Requirements & discovery

Phase 1 · 17:00–17:40 · Full document: [01-understand/brief.md](#)

Problem statement

About 15,000 garment workers — mostly women, many who cannot read well [E-01] — wait at roadside stops in Gazipur for GreenTex staff buses with no idea when their bus will come or whether it will have room. They wait for hours in rain and dark, arrive late, and production stops [E-01]. Boarding is at 6:30 AM in the dark [E-02], in a pushing crowd [E-03], on cheap Android phones with cracked, dim screens and scratched cameras [E-01, E-02], with expensive data and no Wi-Fi at the stop 3 km from the factory [E-01, E-02]. A new cashless 10 Tk fare adds a payment step at the door [E-01].

Product goal. Every worker knows — without needing to read — when to leave home and whether her bus is coming, so she waits minutes, not hours. Supporting goals: board and pay in seconds without a camera; a deliberate, discreet SOS that reaches factory security; works on the real phones; presentable to buyers in English without changing the worker experience.

Who it is for

Tier	User	Source
Primary	Staff-bus worker — mostly women, limited reading, cheap Android, dawn commute	[E-01–03]
Secondary	Factory security-gate officer — receives the SOS with bus number	[E-03]
Tertiary	Bus driver — boards workers; subject of "bus full" and speeding	[E-03, E-04]
Stakeholders	Anisur (operations, buyer demo), Monira (HR), transport managers (future dashboard)	[E-01, E-03]

Functional requirements

ID	Requirement	Source	Priority
FR-1	Personalised home: her route, stop and the next bus for her today — nothing to search	[E-01, E-03]	Must
FR-2	Status as number + words: minutes away · Full — take the next one · Arrived · Delayed / no update since hh:mm ; never show a cached ETA as live	[E-03]	Must
FR-3	Alerts: leave-home cue, bus full, delay — push; SMS fallback proposed [A]	[E-01, E-03]	Must

ID	Requirement	Source	Priority
FR-4	Board and pay: type the daily 4-digit door code (≤ 2 taps + 4 digits), optional QR scan with torch; offline → Pending, Paid 10 Tk only after server confirmation; one transaction ID across retries	[E-01, E-03, E-02]	Must
FR-5	Balance: last-confirmed Tk, pending separate, low-balance warning, recent payments	[E-01]	Must
FR-6	SOS: large one-handed control on waiting and in-transit screens; 5-second countdown with Cancel; states Sending / Not sent / Delivered / Acknowledged ; payload = bus, worker, time, location	[E-01, E-03]	Must
FR-7	Optional Map / Stops one tap from Home: bus and stop on an attributed OpenStreetMap basemap; last-known position when offline	[E-02, E-03]	Must exist, optional use
FR-8	Language: English pre-selected; immediate, prominent বাংলা switch on first launch and every core screen	[E-01]	Must
FR-9	In-transit state: current bus, next stop, ETA to factory, SOS	Rubric	Must
FR-11	Sign-in with company-issued worker ID + password before personalised Home; help and sign-out; no password at each boarding	[D] DL-22	Should
FR-12	Speeding detection: vehicle telemetry + manager review — documented, deferred	[E-04]	Deferred
OUT	Transport-manager dashboard, in-ride groceries	[E-01] "not now"	Out

Non-functional requirements

ID	Requirement	Source
NFR-1	Legibility on dim, cracked screens: primary status contrast $\geq 7:1$, hero digits ≥ 64 px, body $\geq 15\text{--}18$ px, no thin weights	[E-02]
NFR-2	One-handed in a crowd: primary actions in the bottom thumb zone, targets ≥ 56 px with ≥ 12 px gaps, tap-only — no swipes	[E-02, E-03]
NFR-3	Wet-screen tolerance: forgiving taps, clear code feedback, explicit SOS cancel	[E-02]
NFR-4	Connectivity: cache shell/route/identity; small live payloads; every cached status is dated	[E-01, E-02]
NFR-5	Low-end Android: fast cold start; animation only for state change; 360×800 base reflowing $320\text{--}430$ px	[E-01, E-02]
NFR-6	Low-literacy comprehension: ≤ 3 words per state label; number + icon + text + colour; identical layout every day; Bangla-capable typeface	[E-01, E-03]
NFR-7	Safety and trust: SOS reachable at zero balance or offline; a cancelled request is logged, not deleted	[E-01, E-03]
NFR-8	Data cost well under 1 MB/day [A target]	[E-01]
NFR-9	Presentable in English for the buyer demo with one language switch	[E-01]

Contradictions in the client pack — named and resolved

#	Client said	But	Resolution [D]
C1	Live map "exactly like Uber", "absolute must" [E-01, E-02]	Workers "do not understand live maps... just want big text" [E-03]	Status first. Home is a departure board; Map/Stops is one tap away for those who want it
C2	Pay by scanning a complex QR on the door [E-01]	Half the cameras are scratched; 6:30, dark, pushing crowd [E-02, E-03]	Typed 4-digit door code first, QR kept as optional input to the same payment
C3	SOS emails the local police [E-01]	Police "never check it"; alert the factory gate with the bus number [E-03]	SOS goes to factory security with bus, worker, time, location; delivery and acknowledgement shown separately

#	Client said	But	Resolution [D]
C4	English default for the buyer demo [E-01]	Many cannot read properly [E-01]	English stays pre-selected; unmissable বাংলা toggle on every core screen; language-light UI (numbers + icons)
C5	Load the app on factory Wi-Fi [E-01]	The stop is 3 km away; Wi-Fi does not reach; data is expensive [E-01, E-02]	Offline-first dated cache; small updates; SMS fallback proposed [A]
C6	"Premium, smooth animations" [E-01]	Cheap, dim, wet, cracked phones [E-02]	Premium = clarity: one high-contrast light theme, big type, motion only for state change
C7	Dashboard, groceries, speeding [E-01, E-04]	Six-hour sprint; worker app	Parked with IA room; speeding proposed manager-side, no worker UI
C8	15,000 workers / 40 buses [E-01, E-02]	"Bus is full, take the next one" [E-03]	Full is a first-class state because HR names it — not because of fleet arithmetic
C9	Each worker has an "assigned bus" [E-01]	"Take the next one" [E-03]	Route/stop-based assignment: show the next eligible bus on her route
C10	No cash, pay from balance [E-01]	Lateness is "the main thing"; no top-up policy in the pack	Warn before shortfall; grace admission is a client decision, flagged as open

User stories and acceptance criteria

#	Story	Done when
US-1	As a worker, I want to see in one glance how many minutes until my bus, so that I leave home at the right time and do not wait in the rain.	Home shows minutes ≥ 64 px with route and stop; status readable at 7:1 contrast; freshness time visible.
US-2	As a worker, I want to know when my bus is full, so that I wait for the next one instead of pushing.	"Full — next bus in N min" state exists with amber, icon and words; next-bus ETA shown only when known.
US-3	As a worker, I want to board and pay 10 Tk by typing four digits, so that I do not need a working camera in a crowd.	Code entry ≤ 2 taps + 4 digits; wrong code gives retry and help; ticket shows Paid or Pending honestly.
US-4	As a worker travelling at dawn, I want to alert factory security with one hand, so that someone who can act knows my bus number.	SOS ≥ 56 px on Home, ticket and transit; 5 s countdown with Cancel; states Sending → Delivered → Acknowledged.
US-5	As a security officer, I want each SOS to carry bus number, worker, time and location, so that I can respond without calling back first.	Delivered screen lists all four fields and an alert ID; cancellation is timestamped, not erased.

Rough user profiles (Phase 1 hypotheses — checked in Phase 2)

- **Profile W · the worker** [I]: woman on a sewing line, boards ~6:30 in the dark, cheap Android with a cracked screen, limited reading, one hand free in a crowd, pays 10 Tk from a balance she cannot top up in cash.
- **Profile S · the security-gate officer** [I from E-03]: on duty during boarding, needs bus number and location to act; staffing at 6:30 assumed [A].

2 · Research & validation

Phase 2 · 17:40–18:20 · Full document: [02-research/research.md](#)

Method. The client pack (four items) is the only user evidence available in a timed contest. Every fact below is tagged to its source; all testimony is *about* workers, not from them. No interviews, statistics or quotes were invented.

What the scenario taught us

#	Reported fact	Source	What it means for design
F2	Workers "stand in the rain for hours... get late, production stops"	[E-01]	The pain is uncertainty while exposed; the fix is <i>when to leave</i> , not <i>where the bus is</i>
F5	Cheap Android, broken screens, dim brightness	[E-01, E-02]	Contrast, size and forgiving targets outrank visual polish
F6	Data expensive; Wi-Fi does not reach the stop	[E-01, E-02]	Design for interrupted data; date every cached status
F7	"Many cannot read properly, mostly female workers"	[E-01]	Numbers, icons and ≤ 3-word labels carry meaning; Bangla one tap away
F8	"They just want... 'Bus is 10 minutes away' or 'Bus is full, take the next one'"	[E-03]	HR names the two core states almost verbatim → Home is a status board
F9	QR "in a pushing crowd won't work"; half the cameras scratched	[E-03, E-02]	Typed code first; camera optional
F11	HR proposes a 4-digit PIN to confirm boarding	[E-03]	Four digits is plausible; it is a service code, not proof of presence
F12	Police email "useless"; alert the gate "with the specific bus number"	[E-03]	Recipient and payload defined by the person closest to workers
F13	"If they feel unsafe" at dawn → big red SOS	[E-01]	Sponsor acknowledges a safety risk; SOS must be discreet and cancellable
F14	10 Tk daily, no cash, from a balance	[E-01]	Top-up, zero balance and return trips are unspecified → open questions, not invented rules

Main problems (ranked)

Rank	Problem	Severity	Where it lives	Priority
1	Uncertainty while exposed — no arrival / full information in rain and dark [E-01, E-03]	High	Waiting state	P0

Rank	Problem	Severity	Where it lives	Priority
2	Boarding/payment friction — QR in a crowd on damaged cameras; refusal worsens lateness [E-02, E-03]	High	Core boarding flow	P0
3	Comprehension barrier — limited reading, English default, maps [E-01, E-03]	High	Cross-cutting	P0
4	Safety escalation risk — police email rejected; gate recipient proposed [E-01, E-03]	Very high	SOS state	P0
5	Connectivity at the moment of need — no Wi-Fi, costly data [E-01, E-02]	High	Enabler for 1, 2, 4	P0

Frequency is unmeasured for all five; the pack gives no incident rates [A].

Personas — updated from Phase 1

Shahida · sewing-line worker (primary) · illustrative name [A]

Field	Phase 1 guess	Scenario evidence	Phase 2	Status
Goal	Reach the factory before 8:00	"stand in the rain for hours... get late" [E-01]	Narrower daily decision: <i>when to leave, wait or take the next bus</i>	Sharpened
Main need	Know when to leave without reading a paragraph	"big text... 'Bus is 10 minutes away'" [E-03]	Confirmed almost verbatim	Confirmed
Behaviour	Reads Bangla, uses apps	Limited reading [E-01]; PIN suggestion [E-03]	Four digits plausible; script and habits unknown	Corrected / assumed
Constraint	Cheap Android, wet hands	[E-01, E-02]	Confirmed; zero-balance policy added as unknown	Supported + added
Emotion	—	"if they feel unsafe" [E-01]	Fear at dawn acknowledged by the sponsor	Added
Language	—	English default vs literacy [E-01]	Immediate বাংলা switch becomes a design target	Added

Kamal · factory security-gate officer (secondary) · illustrative name [A]

Field	Phase 1 guess	Scenario evidence	Phase 2	Status
Role	Security at the gate during boarding	"alert our factory security gate" [E-03]	Confirmed as intended SOS receiver; 6:30 staffing assumed	Confirmed / assumed
Need	Respond fast	"with the specific bus number" [E-03]	Bus + worker + time + location + alert ID; acknowledgement separate from delivery	Added detail
Constraint	—	[I]	Needs a receiving channel; not a worker-app screen this sprint	Design scope

Tertiary, noted not personified: the bus driver [E-03, E-04] — no automatic driver alert, no silent tracking.

Journey map — rainy morning, bus 15 minutes late (Shahida, Route 3)

Illustrative simulation [A]; only 6:30 boarding and 8:00 shift come from the pack [E-02].

Time	Stage	What happens	Feeling [I]	Without the app	With GreenCommute [D]
05:40	Wake, decide	Rain on the roof; does she leave now?	Anxious	Leaves early "to be safe", waits in rain	Home: Bus 12 · 24 min · updated 05:39 → she waits indoors 10 more minutes
06:05	Walk to stop	15-min walk, umbrella, phone in one hand	Hurried	No information	Push alert: Delayed — 15 min late · no update since 06:02 ; status turns slate-blue
06:20	Wait	Crowd grows; first bus rumoured full	Worried, cold	Pushes toward any bus	Home: Full — next bus 12 min ; she steps back
06:38	Board	Bus 14 arrives; driver shows door code 7319	Rushed	QR scan fails on a scratched, wet camera	Types 7 3 1 9 → blue Thursday ticket, Paid 10 Tk (or Pending if offline)
06:40	Ride	Seated, still dark	Relieved	—	In transit: next stop, ETA 07:25, SOS in thumb reach
06:52	Feels unsafe	Man behind her at the back	Afraid	Nothing to do quietly	Taps SOS → 5 s countdown → Delivered to security · Bus 14 — no sound

Time	Stage	What happens	Feeling [I]	Without the app	With GreenCommute [D]
07:30	Arrive	Factory gate	Calm	Late, wet	On time; security acknowledged at 06:53

Assumptions stated as assumptions [A]: walking distance to the stop; that she owns the phone; that mobile data or SMS exists at the stop; that security is staffed at 06:30; how balance is topped up; whether one 10 Tk fee covers the return trip.

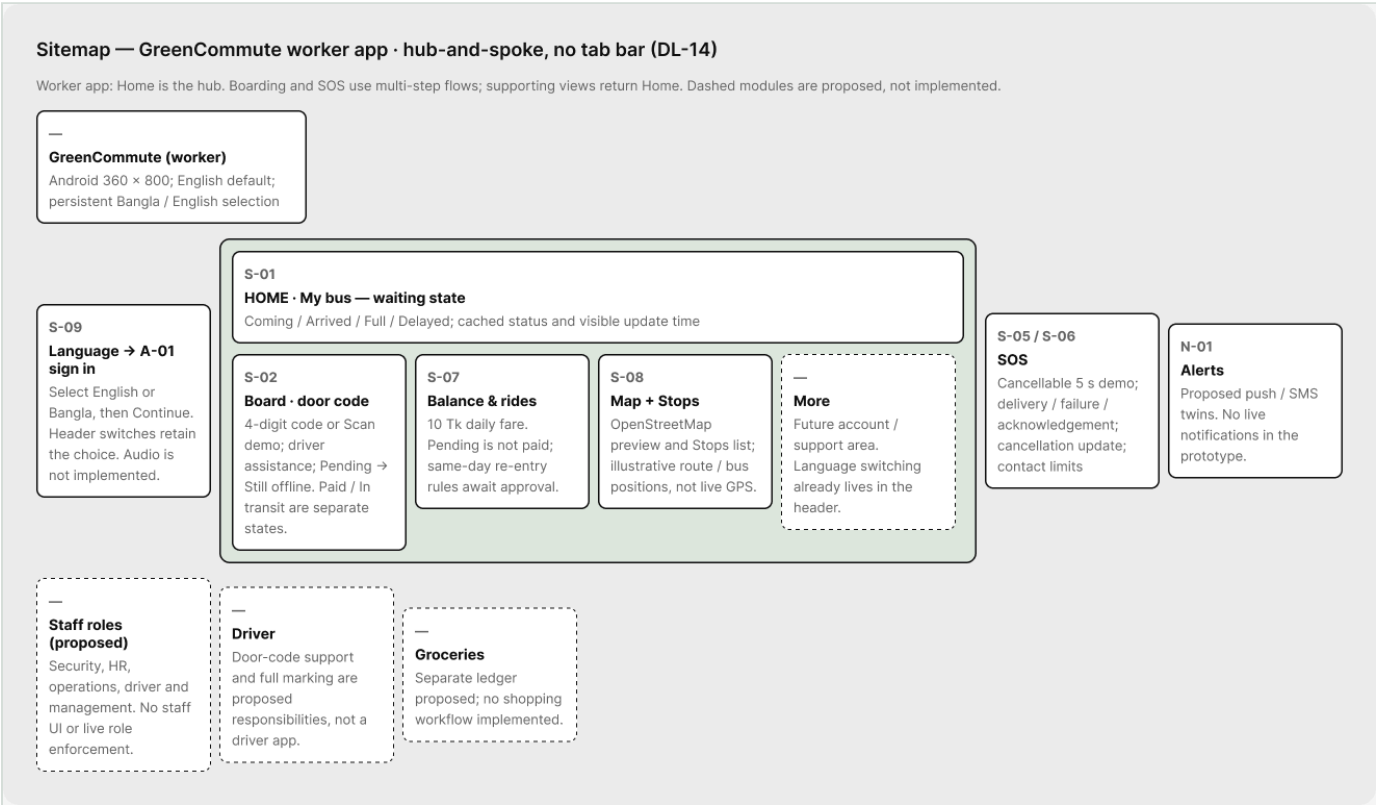
3 · Information architecture & low-fi

Phase 3 · 18:20–19:10 · Full document: [03-structure/structure.md](#) · Figma: [01 Flow & Sitemap](#), [02 Wireframes](#)

Design-system research (adapted, not copied): Material 3 for touch-target and state rules, GOV.UK for plain-language patterns, Uber Base for transit status hierarchy, BBC GEL for legibility. Point of view for the product: *a bus-stop departure board in her hand*.

Sitemap — hub and spoke (DL-14)

Home is the only hub; every spoke is one tap deep and returns to Home. No tab bar, no hamburger.

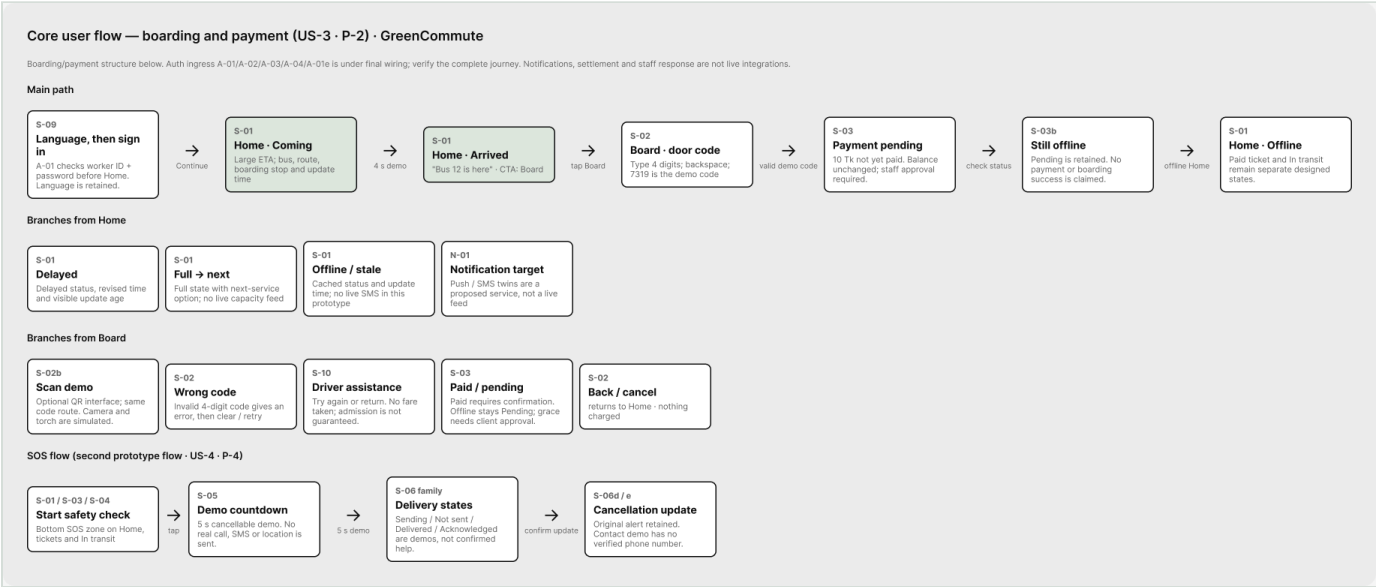


Sitemap · Figma page 01

Screen	Purpose	User story	Primary action
S-09 Language	First launch: English pre-selected, বাংলা one tap	US-1 (C4)	Continue
A-01...A-05 Sign-in	Worker ID + password, help, demo credentials, checking, sign-out	FR-11	Sign in
S-01 Home	Status board: minutes / Full / Arrived / Delayed / Offline; SOS	US-1, US-2	Board · View map · SOS
S-02 Door code	Type 4 digits, or Scan instead (S-02b torch scanner); wrong-code retry; driver help (S-02c)	US-3	Pay 10 Tk
S-03 Ticket	Day-colour ticket: Paid / Pending (still offline S-03b)	US-3	Show driver · SOS

Screen	Purpose	User story	Primary action
S-04 In transit	Bus, next stop, ETA to factory, SOS	FR-9, US-4	SOS
S-05 SOS countdown	5 s countdown, Cancel	US-4	Cancel
S-06 SOS states	Sending · Delivered · Not sent · Acknowledged · Confirm cancellation · Cancellation logged · No contact	US-4, US-5	Call security · Return
S-07 Balance	Confirmed Tk, pending, recent payments, sign-out	FR-5	Back
S-08 Map / Stops	Optional geographic map + stop list	FR-7 (C1)	Home

Core flow — boarding and payment



Core flow: boarding + payment with wrong-code, offline and help branches · Figma page 01

Worker account access

[D] User-authorized addition. Research: NIST SP 800-63B-4, WCAG 2.2 SC 3.3.8, Android Credential Manager, OWASP Authentication. No worker interviews or live identity service.

Entry: S-09 -> A-01 -> A-04 -> S-01

Language → worker ID + password → simulated check → existing Home only on success. A valid personal-device session may avoid repeat entry in the proposed production service.

Denied: empty / incorrect / offline / unavailable / throttled

Empty fields remain on A-01. Incorrect details, no internet, service failure and rate limits show A-01e and never open Home. Credentials are cleared; the entered worker ID is retained for correction.

Recovery: A-01 -> A-02 -> A-01

Get help signing in → proposed HR identity check and secure setup/reset → back to sign in. A worker ID alone cannot reset an account. HR operations and recovery policy remain U-15.

Sign out: S-07 -> A-05 -> A-01

Balance → confirmation → clear authentication state. Cancel returns to Balance. Pending payment ownership is preserved; this prototype does not revoke a real session.

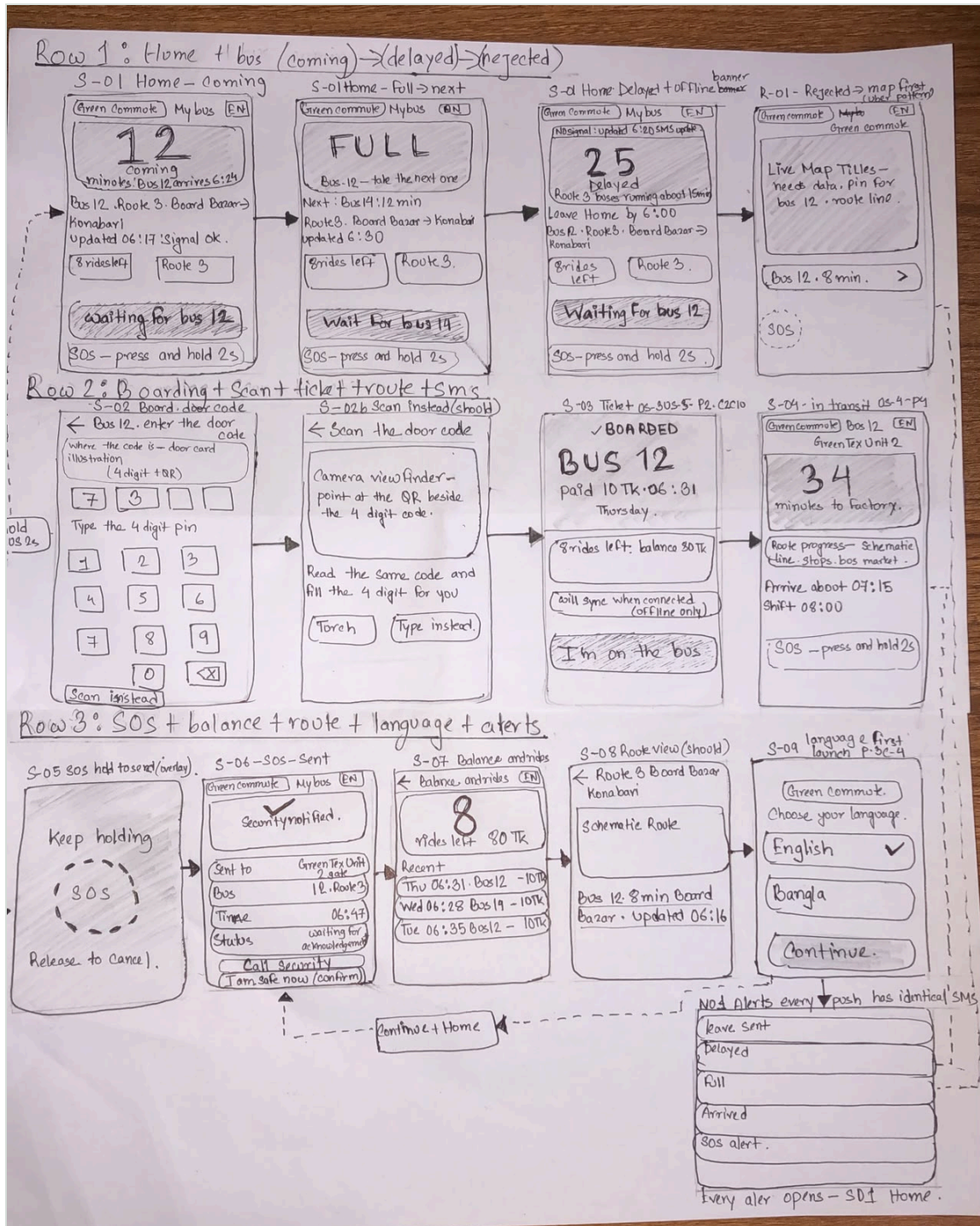
Prototype scope

A-03 is a fixed sample-credential chooser, not Android autofill or a real keyboard. Tester controls simulate responses. English/Bangla persist; boarding code, payment and SOS contracts stay separate.

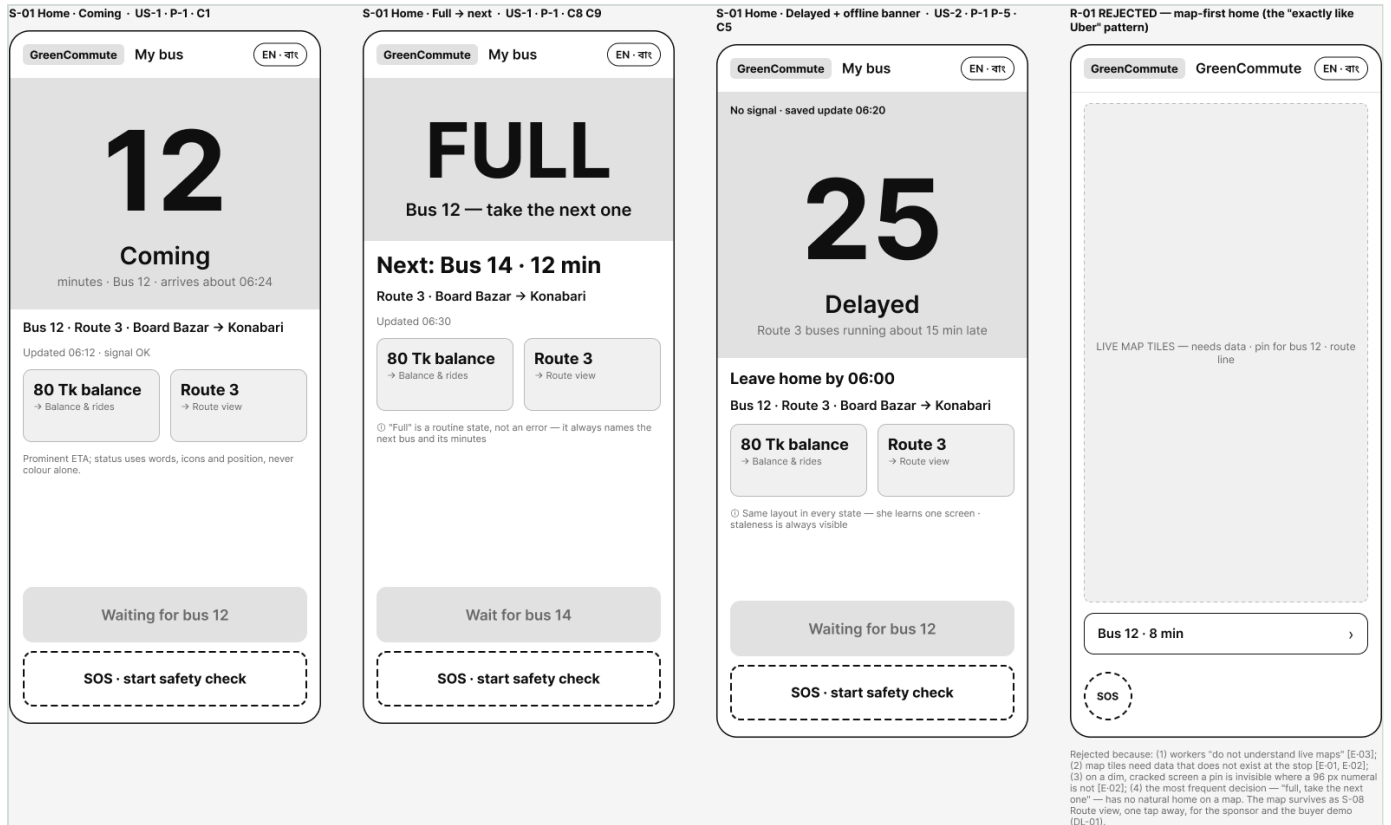
Entry flow: language and sign-in

Wireframes

Hand sketch first, then the Figma low-fi frames for every committed screen.

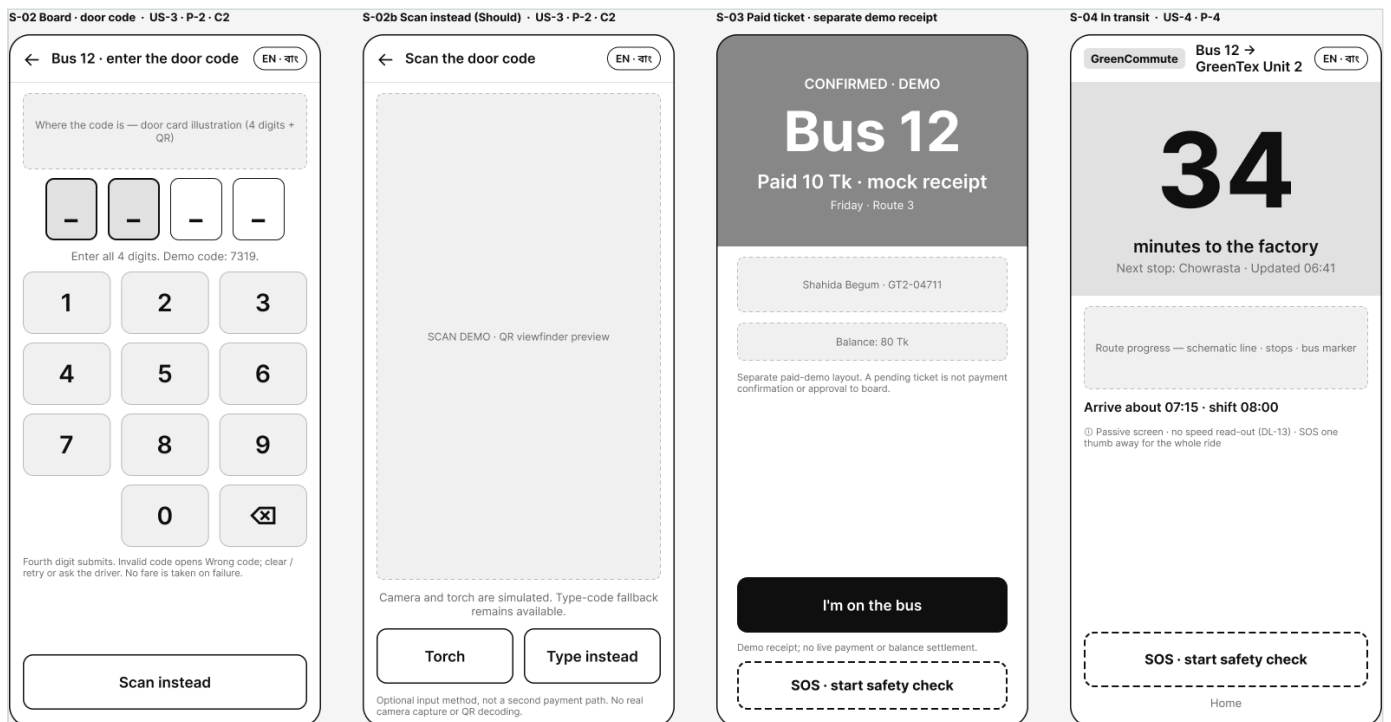


Hand sketch · Mohammed Tayeb · 11 Sep 2026, Phase 3

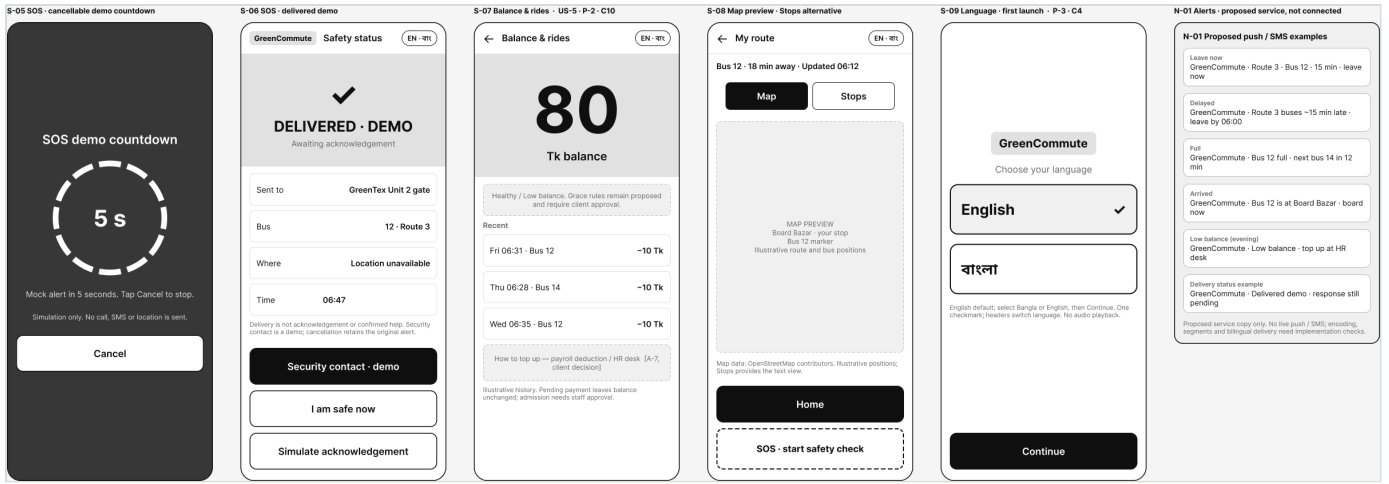


A · Home states + **R-01, the rejected home alternative** (right)

Why R-01 was rejected. R-01 put the live map on top with the ETA as a caption — the client's Uber request [E-01]. It fails Monira's evidence [E-03] (workers do not read maps), spends the top 60 % of a dim screen on tiles that cost data [E-01, E-02], and pushes the SOS below the thumb zone. The chosen S-01 puts minutes at 128 px, status colour behind them, and SOS in the bottom band; the map is one tap away.



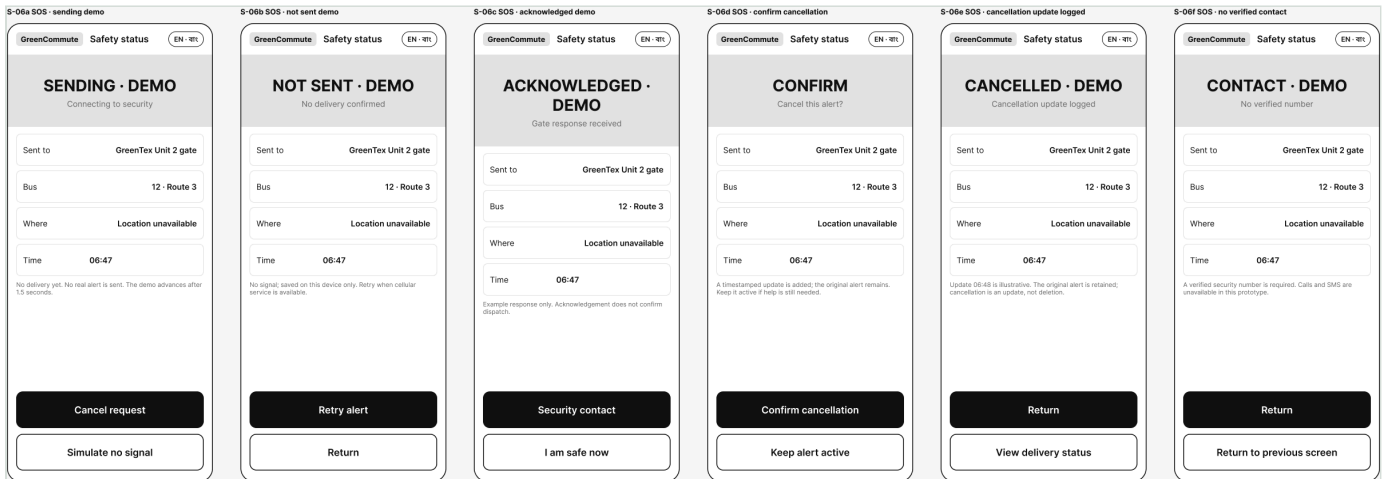
B · Door code, scanner, ticket, in transit



C · SOS, balance, route, language, alerts



D · Recovery states and stops



E · SOS recovery states

A-01 Worker sign in - late structural reconciliation

Historical structure: bands A-C are Phase 3 sketches; D-F are late reconciliation, not original research. Current delta: tap SOS → 5 s countdown, Cancel → Back; funds are Tk, not ride counts. Daily fare is 10 Tk; same-day policy remains open.

Back

EN / Bangla

Your worker account

Worker ID

Password - masked

Keep me signed in
Only on my own phone

Use saved sign-in

Sign in

Get help signing in

Prototype: sample account only

A-01 remains here: low-fi 169:3790 / source 164:1944. Five supporting auth sketches are now grouped on adjacent board 211:1815: help 169:4037; chooser 169:4041; checking 169:4045; result 169:4049; sign out 204:1681. Late QA reconciliation · 11 Sep 2026; added after implementation. US-7 · DL-22.

Review after auth-owner wiring: language choice → sign-in → sample credentials/checking → result → requested task. Check help, Back, invalid/offline outcomes and locale retention. No production authentication or successful replay is asserted.

F · Sign-in states

4 · High-fidelity UI

Phase 4 · 19:10–20:40 · Notes: [04-screens/screen-notes.md](#) · Figma: [03 Screens](#)

Designed for dim screens and one hand. One light, high-contrast theme (every primary text pair $\geq 7:1$, measured in \$5); hero digits 128 px; all primary controls ≥ 56 px in the bottom thumb band; tap-only; red reserved for SOS; identical layout every day so she learns it once. Android 360 × 800 base; auto-layout reflows 320–430 px.

Waiting state — S-01 Home

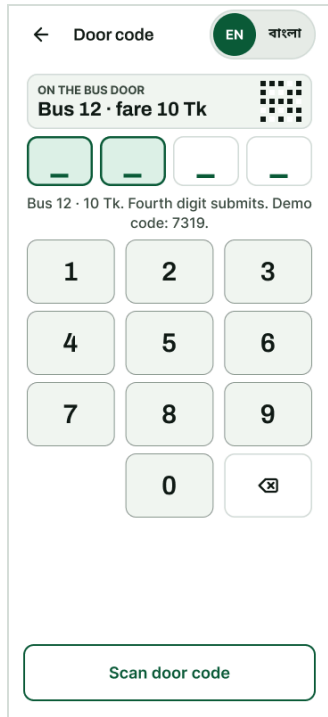
Coming · US-1, F8 · minutes at 128 px, updated time, Board + View map, SOS in thumb band

Arrived · US-1 · green fill, "Board now" becomes the single big action

Full · US-2, C8 · amber, icon + words, next-bus ETA only when known

Offline · C5, Problem 5 · cached status dated, never shown as live; SOS still present

Boarding and payment — S-02 → S-03



← Door code EN বাংলা

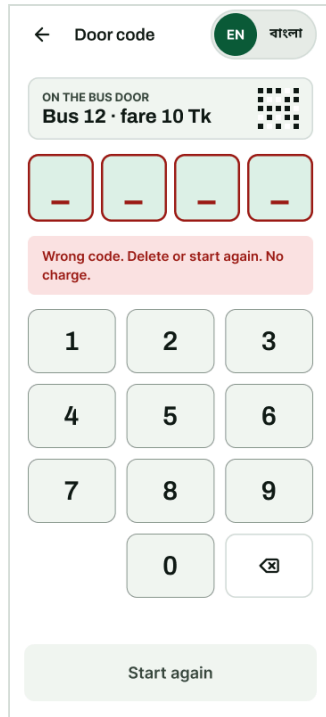
ON THE BUS DOOR
Bus 12 · fare 10 Tk

Bus 12 · 10 Tk. Fourth digit submits. Demo code: 7319.

1 2 3
4 5 6
7 8 9
0

Scan door code

Door code · US-3, C2 · 4 slots, 56 px keys, camera optional



← Door code EN বাংলা

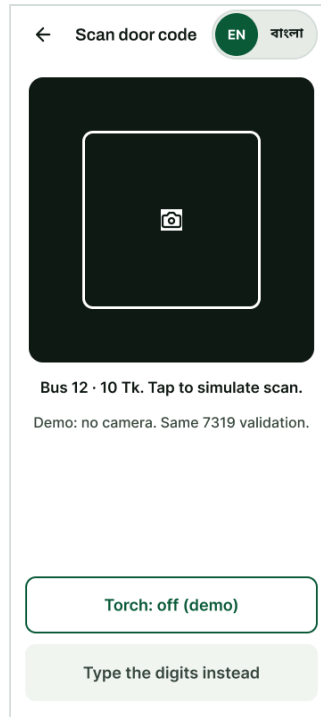
ON THE BUS DOOR
Bus 12 · fare 10 Tk

Wrong code. Delete or start again. No charge.

1 2 3
4 5 6
7 8 9
0

Start again

Wrong code · Problem 2 · plain words, retry, help after 3 failures



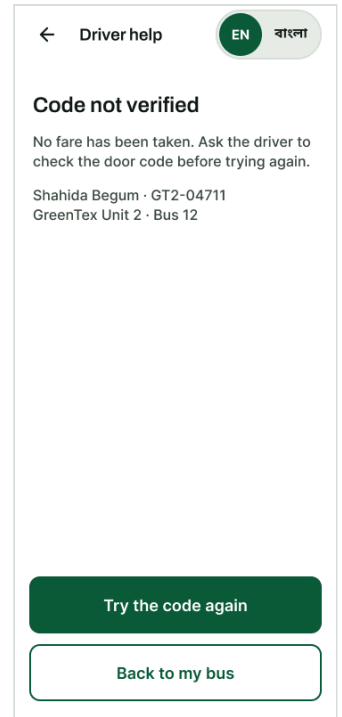
← Scan door code EN বাংলা

Bus 12 · 10 Tk. Tap to simulate scan.
Demo: no camera. Same 7319 validation.

Torch: off (demo)

Type the digits instead

Scan instead · C2 · torch for 6:30 dark; fills the same four digits



← Driver help EN বাংলা

Code not verified

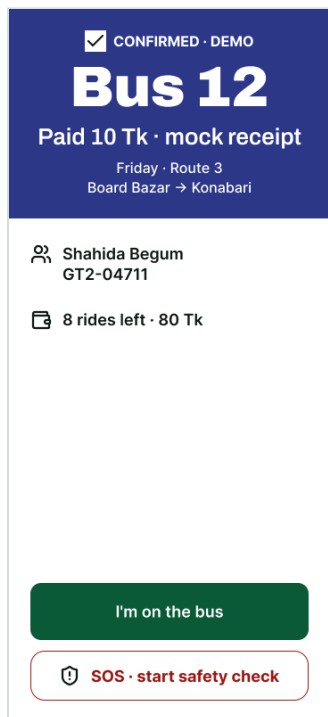
No fare has been taken. Ask the driver to check the door code before trying again.

Shahida Begum · GT2-04711
GreenTex Unit 2 · Bus 12

Try the code again

Back to my bus

Driver help · US-3 · what to show the driver; no admission promise



CONFIRMED · DEMO

Bus 12

Paid 10 Tk · mock receipt

Friday · Route 3
Board Bazar → Konabari

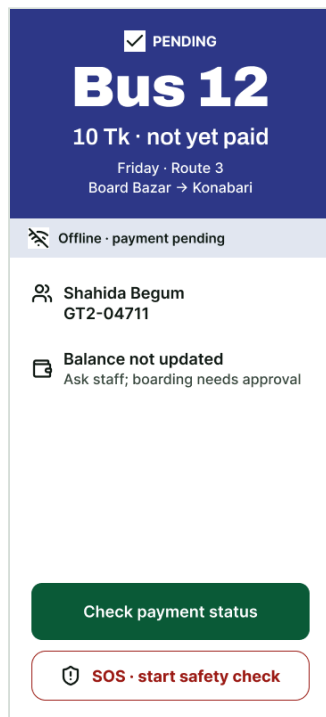
Shahida Begum
GT2-04711

8 rides left · 80 Tk

I'm on the bus

SOS · start safety check

Ticket · Paid · US-3 · day colour + big bus number the driver reads at arm's length



PENDING

Bus 12

10 Tk · not yet paid

Friday · Route 3
Board Bazar → Konabari

Offline · payment pending

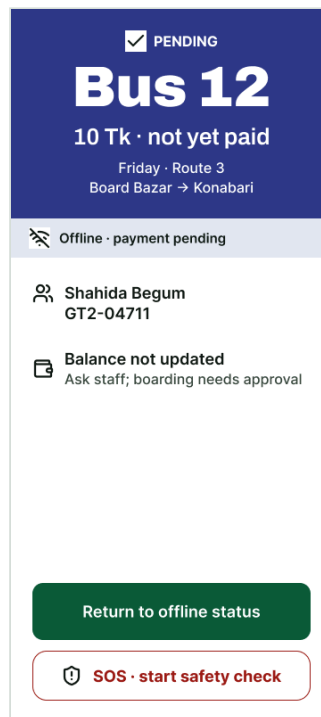
Shahida Begum
GT2-04711

Balance not updated
Ask staff; boarding needs approval

Check payment status

SOS · start safety check

Ticket · Pending · C5, C10 · honest: paid only after server confirms



PENDING

Bus 12

10 Tk · not yet paid

Friday · Route 3
Board Bazar → Konabari

Offline · payment pending

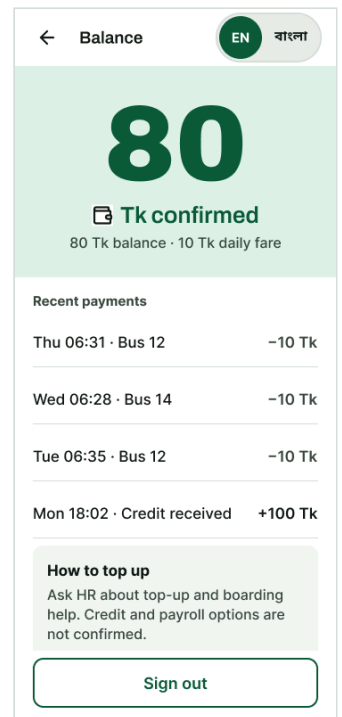
Shahida Begum
GT2-04711

Balance not updated
Ask staff; boarding needs approval

Return to offline status

SOS · start safety check

Still offline · Problem 5 · pending stays pending; no invented success



← Balance EN বাংলা

80

Tk confirmed
80 Tk balance · 10 Tk daily fare

Recent payments

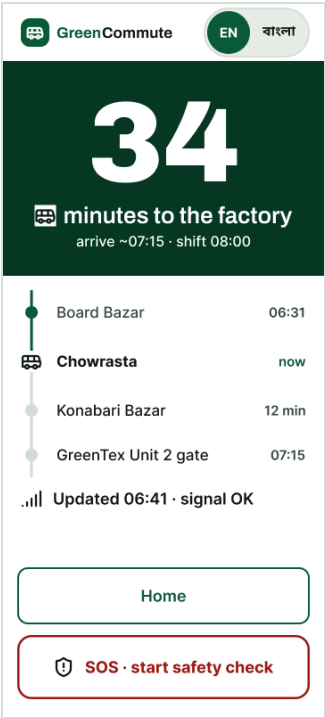
Thu 06:31 · Bus 12	-10 Tk
Wed 06:28 · Bus 14	-10 Tk
Tue 06:35 · Bus 12	-10 Tk
Mon 18:02 · Credit received	+100 Tk

How to top up
Ask HR about top-up and boarding help. Credit and payroll options are not confirmed.

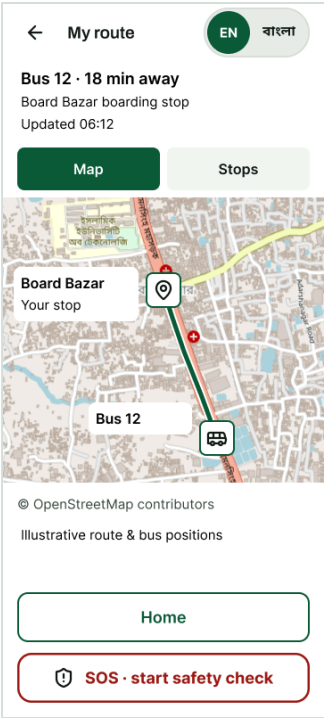
Sign out

Balance · FR-5 · confirmed vs pending Tk, recent payments, sign-out

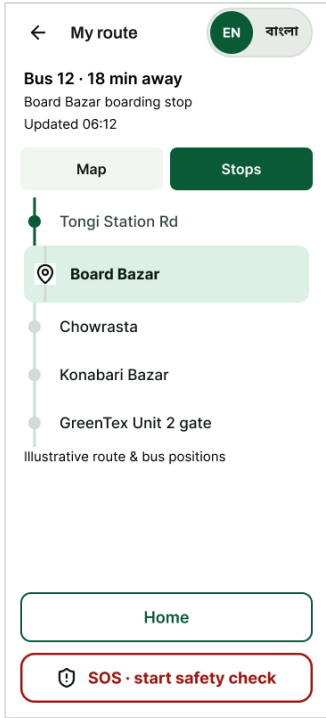
In-transit state — S-04, S-08



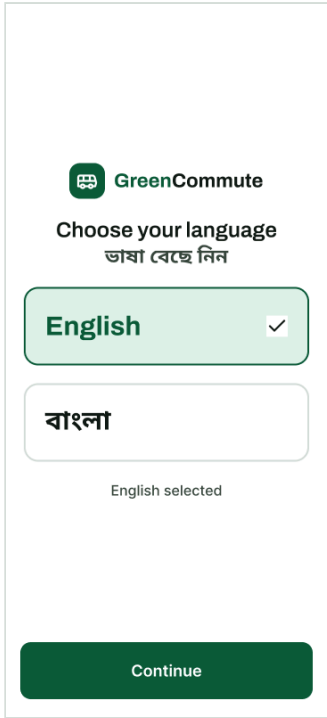
In transit · FR-9, US-4 · next stop, ETA to factory, SOS in thumb reach



Map · C1, FR-7 · optional; attributed OSM tiles; position timestamp

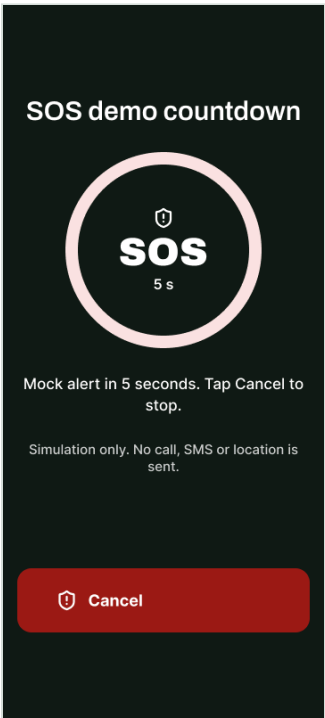


Stops · C1 · schematic alternative for map-averse users

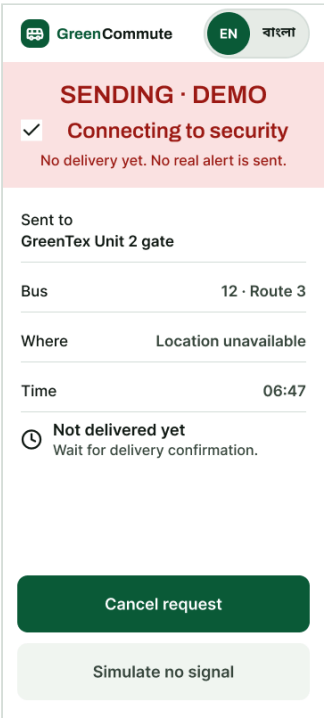


Language · C4 · English pre-selected, বাংলা one tap; whole app switches

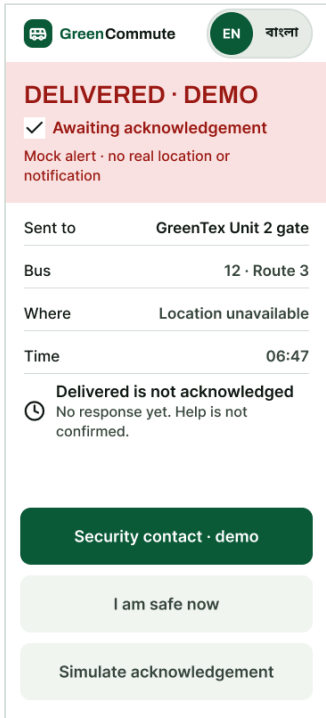
Panic / SOS state — S-05, S-06



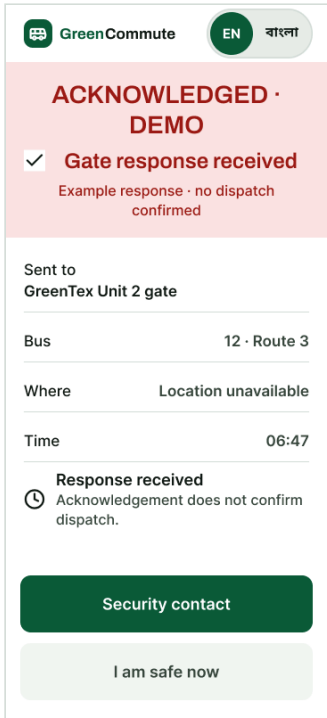
Countdown · US-4, C3 · 5 s to cancel an accidental press; no sound



Sending · NFR-7 · attempt under way, Cancel still available



Delivered · US-5 · bus 14, worker, time, location, alert ID; Call security



Acknowledged · US-5 · someone is acting — distinct from delivered

Not sent · C5 · no channel → say so, retry, call

Confirm cancel · NFR-7 · cancel after delivery needs confirmation

Cancellation logged · US-5 · timestamped, not erased

No contact · [A] U-7 · number unknown; no fabricated phone number

Entry — sign-in (A-01...A-05, DL-22)

Sign-in

Help

Demo credentials

Checking

Result

Sign-out

Company-issued worker ID + password once per device, never at each boarding (NFR-3). Sample credentials are labelled as demo; failures use generic wording; help does not reset by ID alone.

5 · Figma ecosystem & design file

Phase 5 · 20:40–21:50 · Full document: [05-design-file/design-system-and-accessibility.md](#) · Figma: [04 Design System, 05 Accessibility](#)

Named variables

Collection **GreenCommute tokens** — semantic colour variables aliased to primitives; collection **GreenCommute language** — `locale` mode (English / Bangla) binding 153 copy variables plus font family/style so every screen switches language and typeface from one toggle.

Group	Variables	Value	Why
Surface	<code>surface/page</code> · <code>surface/alt</code> · <code>surface/border</code>	<code>#FFFFFF</code> · <code>#F3F6F4</code> · <code>#D5DDD8</code>	One light theme: readable outdoors through droplets and cracks (DL-11)
Text	<code>text/primary</code> · <code>text/secondary</code> · <code>text/brand</code> · <code>text/on-color</code>	<code>#0F1A14</code> · <code>#3A4A40</code> · <code>#0B5D3B</code> · <code>#FFFFFF</code>	Near-black, never grey, on dim LCDs
Brand	<code>brand/primary</code> · <code>brand/tint</code> · <code>brand/deep</code>	<code>#0B5D3B</code> · <code>#DDF3E7</code> · <code>#063B26</code>	Green from "GreenCommute" / GreenTex; deep enough to carry white text
Status	<code>status/coming</code> · <code>status/full</code> · <code>status/delayed</code> (bg + fg each)	green <code>#DDF3E7</code> / <code>#063B26</code> · amber <code>#FFE8B8</code> / <code>#5E3D00</code> · slate <code>#E3E9F2</code> / <code>#2B4A73</code>	Three states HR named; each has icon + words, never colour alone
Danger	<code>danger/bg</code> · <code>danger/tint</code> · <code>danger/fg</code> · <code>danger/text</code>	<code>#9E1B14</code> · <code>#FBE3E1</code> · <code>#FFFFFF</code> · <code>#9E1B14</code>	Red reserved for SOS so it stays meaningful
Ticket	<code>ticket/thursday</code>	<code>#2E3A87</code>	Day colour of the boarding ticket (rotates daily, DL-02)
Language	<code>`toggle/en</code>	<code>bn/bg</code>	<code>fg`</code>

Font set

Font	Roles	Why
Archivo Black / SemiBold / Bold	<code>display/*</code> 36–128 px (hero minutes, bus numbers, balance) · <code>heading/*</code> 17–30 px	Wide, heavy grotesk reads like a departure board at arm's length; tabular-feeling digits
Inter Regular–Bold	<code>body/*</code> 15–18 px · <code>label/*</code> 13–18 px · <code>caption</code> 15 px	Tall x-height for small dim displays
Noto Sans Bengali Regular–Black	<code>bangla/*</code> , bound to the same roles in Bangla mode	Correct conjunct shaping; keeps English size and weight when switching

All three are Google Fonts under the SIL Open Font License.

Components with variants · auto layout

- **Button** set: Kind (Primary / Secondary / Muted / Danger) × State (Default / Pressed / Disabled / Focus) = 16 variants, 56–72 px tall.
- **Status hero** (Coming / Full / Arrived / Delayed / Offline), **Key** (digit / delete / confirm), **Banner** (info / warning), **Language chip** (segmented EN / বাংলা, 116 × 56), **SOS bar**, **Top bar**, **Detail row**, **Ticket card**.
- Every screen frame is auto layout with Fill/Hug sizing; verified at 320 / 360 / 412 / 430 px widths and 640 / 800 heights in both languages with zero text overflow. Layers are named by role (e.g. `Status hero`, `Primary button`, `SOS bar`).
- **Real text everywhere** — no lorem ipsum; sample names, codes and times are marked illustrative. **Theme logo**: the pack supplies only a name, so an original GreenCommute wordmark (Archivo Black, brand green, leaf-bus mark) is used on Language, Sign-in and Home.

Clickable prototype

Two flow starts on page 03 Screens: **Boarding** (Language → Sign-in → Home → Door code → Ticket → In transit, with wrong-code, scan, offline and help branches) and **SOS trigger** (In transit → SOS → 5 s countdown → Sending → Delivered → Acknowledged / Cancel). Every visible control is wired; Back uses navigation history; timers use `After delay`. Links on the cover.

Accessibility notes — written down, not just claimed

Text	On	Ratio	WCAG
text/primary #0F1A14	surface/page #FFFFFF	17.8 : 1	AAA
text/secondary #3A4A40	surface/page	9.4 : 1	AAA
text/brand #0B5D3B	surface/page	7.95 : 1	AAA
text/on-color #FFFFFF	brand/primary #0B5D3B (buttons, Arrived)	7.95 : 1	AAA
status/coming/fg #063B26	status/coming/bg #DDF3E7 (128 px hero)	10.9 : 1	AAA
status/full/fg #5E3D00	status/full/bg #FFE8B8	8.1 : 1	AAA — deepened from 4.9:1 in audit
status/delayed/fg #2B4A73	status/delayed/bg #E3E9F2	7.4 : 1	AAA
danger/fg #FFFFFF	danger/bg #9E1B14 (SOS bar)	8.0 : 1	AAA
danger/text #9E1B14	danger/tint #FBE3E1	6.55 : 1	AA (below our internal 7:1 — noted)
icon/muted #8A968F	surface/page	3.1 : 1	AA non-text — decorative only

- **Alt text**: every image and icon in this document carries alt text; in Figma each screen frame has a description and icons are named by meaning (`icon/bus`, `icon/warning`) for developer-side content descriptions.
- **Not colour alone**: every status has icon + ≤ 3 words + colour; ticket day-colour is backed by the day name.
- **Touch and motion**: targets ≥ 56 px with ≥ 12 px gaps; tap-only; animation only on state change; SOS is silent and cancellable for 5 s.

- **Language:** whole-app Bangla with Noto Sans Bengali; numbers stay large and Western-digit for driver recognition [A — to test].
- **Focus:** Button Focus variant with 3 px brand/deep ring for keyboard/switch access.
- **Not verified:** real-device TalkBack, worker comprehension tests, Bangla translation review — listed as open questions, not claimed.

6 • Handoff

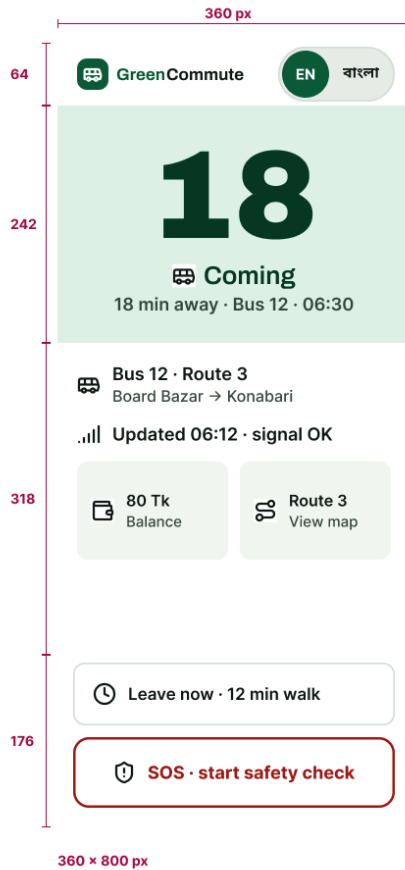
Phase 6 · 21:50–22:40 · Full note: [06-handoff/handoff.md](#) · Figma: [06 Prototype](#), [07 Redlines & Handoff](#)

Redlines — home screen (S-01)

Redlines · S-01 Home · Coming

360 × 800 px · Current Home reference from 03 Screens / S-01 (27:9)

Padding order: top / right / bottom / left · Annotation copy is not an interactive journey



Top bar · 360 × 64 px x 0 · y 0-64

Component Top bar / Kind=Home
Pad 0/16/0/20 · gap 0
horizontal · FILL × FIXED
Fill surface/page
Type label/bold-15, bangla/label, heading/bold-17
Text text/primary, text/brand

Brand mark · 171 × 32 px x 20 · y 16-48

Component Mark / GreenCommute · compact
Pad 0/0/0/0 · gap 8
horizontal · HUG × HUG
Fill action/primary/fg
Type heading/bold-17
Text text/brand, text/primary

Language chip · 122 × 56 px x 222 · y 4-60

Component Language chip
Pad 12/14/12/14 · gap 6
horizontal · HUG × FIXED
Fill surface/page
Radius 20
Type label/bold-15, bangla/label
Text text/primary, text/brand

Status block · 360 × 242 px x 0 · y 64-306

Component Status block / State=Coming
Pad 28/20/28/20 · gap 2
vertical · FILL × HUG
Fill status/coming/bg
Type display/xxl, label/lg, heading/lg-26
Text status/coming/fg, text/secondary

Content · 360 × 318 px x 0 · y 306-624

Pad 20/20/12/20 · gap 16
vertical · FILL × FILL
Type label/lg, body/medium-16, label/md, caption
Text text/primary, text/secondary

Bus detail · 320 × 45 px x 20 · y 326-371

Component Detail row
Pad 0/0/0/0 · gap 12
horizontal · FILL × HUG
Type label/lg, body/medium-16
Text text/primary, text/secondary

Detail text · 284 × 45 px x 56 · y 326-371

Pad 0/0/0/0 · gap 0
vertical · FILL × HUG
Type label/lg, body/medium-16
Text text/primary, text/secondary

Tiles row · 320 × 100 px x 20 · y 427-527

Pad 0/0/0/0 · gap 12
horizontal · FILL × HUG
Type label/md, caption
Text text/primary, text/secondary

Balance tile · 154 × 100 px x 20 · y 427-527

Component Tile / Kind=Balance
Pad 14/12/14/14 · gap 12
horizontal · FILL × FIXED
Fill surface/alt
Radius 12
Type label/md, caption
Text text/primary, text/secondary

Route tile · 154 × 100 px x 186 · y 427-527

Component Tile / Kind=Route
Pad 14/12/14/14 · gap 12
horizontal · FILL × FIXED
Fill surface/alt
Radius 12
Type label/md, caption
Text text/primary, text/secondary

Bottom zone · 360 × 176 px x 0 · y 624-800

Pad 8/16/20/16 · gap 12
vertical · FILL × HUG
Type label/semi-17, label/bold-18
Text text/primary, danger/text

Guidance · 328 × 64 px x 16 · y 632-696

Component Guidance banner
Pad 0/20/0/20 · gap 12
horizontal · FILL × FIXED
Radius 12
Type label/semi-17
Text text/primary

SOS bar · 328 × 72 px x 16 · y 708-780

Component SOS bar / State=Idle
Pad 0/0/0/0 · gap 12
horizontal · FILL × FIXED
Fill surface/page
Radius 16
Type label/bold-18
Text danger/text

Tokens & behaviour

Layout and touch targets

Size tokens
size/target 56; size/cta 64
size/key 72; size/sos 72

Current Home dimensions
Guidance 328 × 64; SOS 328 × 72
Tiles 154 × 100; Language 122 × 56

Keypad: fill available width/height; 98.7 × 72 at 360 × 800, not a fixed key width. Tickets: 24 px gutters, 64 px bus heading, 56 px SOS. Delivered SOS: left-aligned 28/18/16 px. Native targets require device verification.

Spacing

Colour values

surface/page #FFFFFF
surface/alt #F3F6F4
surface/border #D5DDD8
text/primary #0F1A14
text/secondary #3A4A40
text/on-color #FFFFFF
text/brand #0B5D3B
brand/primary #0B5D3B
brand/tint #DDF3E7
brand/deep #063B26
status/coming/bg #DDF3E7
status/coming/fg #063B26
status/arrived/bg #0B5D3B
status/arrived/fg #FFFFFF
status/fulfilled/bg #FFFFFF

Type and borders

display/xxl · Archivo Black 128
display/lg · Archivo Black 64
heading/xl · Archivo SemiBold 28
heading/lg-26 · Archivo SemiBold 26
heading/bold-17 · Archivo Bold 17
label/lg · Inter Semi Bold 18
label/md · Inter Semi Bold 16
label/bold-18 · Inter Bold 18
label/bold-15 · Inter Bold 15
label/semi-17 · Inter Semi Bold 17
body/md · Inter Regular 16
body/medium-16 · Inter Medium 16
caption · Inter Medium 15
bangla/label · Noto Sans Bengali Bold 15

space/1 4 · space/2 8 · space/3 12
space/4 16 · space/5 20 · space/6 24
space/8 32 · space/10 40

Radii
radius/sm 8 · radius/md 12
radius/lg 16 · radius/xl 24

status/id/bg #FFC000
status/full/bg #5E3D00
status/delayed/bg #E3E9F2
status/delayed/bg #2B4A73
danger/bg #9E1B14
danger/tint #FBE3E1
danger/text #9E1B14
action/muted/bg #F3F6F4
action/muted/bg #3A4A40
ticket/thursday #2E3A87

Strokes: Home SOS 2 px danger/bg; ticket SOS 1 px.
Language and guidance use surface/border. The
1.38:1 surface border is decorative, not a sole
meaningful boundary. See Accessibility for
composed-colour limits.

Home source 27:9 and annotation copy 169:3943 match: 18 min ETA; Leave now, 12 min walk; source update 06:12. These are illustrative values. Cached positions are not live; the 3 min stale threshold remains proposed. Balance opens S-07; View map/Stops support S-08.

Boarding: enter all four demo digits, 7319. Daily fare is 10 Tk; 80 Tk is a balance, not eight rides. Same-day re-entry rules need approval. Pending is not paid or admission; checking can remain offline. Paid receipt and In transit are separate demo states.

Language/auth: preserve English/Bangla choice and bound fonts. Review A-01 sign-in, A-02 help, A-03 chooser, A-04 checking and A-01e result and A-05 sign-out confirmation (204:1681) after auth wiring. No completed replay is claimed.

Safety: tap → 5 s countdown; Cancel → Back; 1.5 s mock sending. Delivered is not acknowledged; acknowledgement is not dispatch. Cancellation retains the alert. No live calls, SMS, GPS or settlement.

Worker UI only; staff roles and manager-side speeding remain proposed. Approve service policy and native accessibility before a pilot. Route raster credit: <https://www.openstreetmap.org/copyright>

Home redlines · spacing, sizes, tokens and text styles for every element · Figma page 07

Developer handoff note

One page · GreenCommute · Mohammed Tayeb · 11 Sep 2026 · [Design file](#) · [Prototype](#) · Redlines: page 07

Purpose. Android app for Gazipur garment workers: bus arrival status, camera-free 10 Tk boarding, discreet SOS to factory security. Constraints: low-end Android, dawn/rain, limited reading. English default, Bangla one tap.

Main flows

- **Entry:** Language → Sign-in (worker ID + password) → Checking → Home. Failure → Result with generic message; Help does not reset by ID.
- **Board:** Home → Door code (type 4 digits or Scan) → Ticket → In transit. Wrong code: retry; after 3 failures offer Driver help — never admission. Offline: ticket **Pending; Paid** only on server confirmation.
- **SOS:** Home / Ticket / In transit → 5 s countdown (Cancel) → Sending → Delivered → Acknowledged. Not sent when no channel. Cancel after delivery needs confirmation and is logged with time, not deleted.
- **Other:** Home → Balance / Map · Stops → back. Balance → Sign-out confirmation.

Build with

- Tokens: `GreenCommute tokens` (colour) and `GreenCommute language` (locale mode, copy, fonts). Fonts: Archivo, Inter, Noto Sans Bengali (OFL).
- Components: Button (Kind × State), Key, Status hero, Banner, Language chip, SOS bar, Top bar, Detail row, Ticket card. Base 360 × 800; test 320 / 412 / 430 reflow, Back, keyboard, focus, ≥ 56 px targets.
- **Status:** show bus, stop, ETA and source time; cache is dated, never presented as live. Proposed stale threshold 3 min.
- **Payment:** door code identifies the service, not presence or payment. Verify service, date and fare server-side. Typed, QR and retries share one transaction ID — no duplicate debits.
- **SOS:** one alert ID reused across retries; payload = worker, bus, time, location + accuracy + age. Delivered ≠ acknowledged. Never block on GPS. Figma sends nothing.
- **Auth:** company-issued ID; masked password with Show/Hide; paste/autofill allowed; generic errors; server rate limits; TLS; finite revocable sessions; no offline first-login.

Open questions for the client

1. Does one 10 Tk fee cover the return trip? Service-day cutoff? (FARE-1–5)
2. Top-up method and zero-balance policy — grace admission needs approval (C10).
3. Who marks a bus full and how fresh is that signal? (U-4)
4. Security-gate staffing at 06:30, receiving channel and call-back number (U-7).
5. Mobile data / SMS coverage at the stops; SMS fallback cost (U-7).
6. Speeding: telemetry source, thresholds, manager review — deferred by us (FR-12).
7. Bangla translation review and audio prompts; worker comprehension test on real phones.

Not in this prototype: live backend, real payment, GPS, SMS, audio playback, staff dashboard, groceries.

7 · Tools, AI use & credits

Tools and AI use (Rulebook §08)

Phase	Tool	How it was used
1–3	GitHub Copilot (VS Code)	Structured the client pack into requirements, contradictions, personas, IA; challenged weak decisions; drafted the phase documents from my inputs
3	Pen and paper	Hand-sketched wireframes (page 8) before Figma low-fi
4	Figma + Copilot with Figma MCP	Hi-fi screens built from my wireframes and design brief; AI executed layout and variable wiring under review
5	Figma + Copilot with Figma MCP	Variables, variants, auto-layout audit, contrast measurement, text-overflow checks at five widths
6	Figma prototyping; Copilot	Prototype wiring, redlines, handoff note, this PDF
QA	Copilot subagents as independent "judges"	Rubric scoring, first-time-user walkthroughs, file QA — findings fixed in a loop

This eight-prompt pack was the **main workflow**: chatgpt.com/s/t_6aa430b886508191ac12f16790ebc536. Many additional prompts, audits and research passes were run during the sprint — independent judge simulations, Figma structure and prototype QA, accessibility contrast checks, design-system research (Material 3, GOV.UK, Uber Base, BBC GEL). AI acted as an assistant; every design decision was made by me and is traceable, with evidence and alternatives, in the decision log: [DECISIONS.md](#) (DL-01...DL-23) and [design_decision.md](#). No user interviews, statistics or quotes were fabricated; illustrative values are marked [A].

Credits (Rule 04)

- **Fonts:** Archivo, Inter, Noto Sans Bengali — Google Fonts, SIL Open Font License 1.1.
- **Icons:** Lucide — ISC License.
- **Map tiles:** © OpenStreetMap contributors, ODbL — attributed on the Map screen; positions illustrative.
- **Logo:** original GreenCommute wordmark and mark, created during the sprint (the pack supplied only the name).
- **Everything else** — copy, screens, components, diagrams, sketch — original work by Mohammed Tayeb, 11 September 2026, 17:00–23:00.